# Intro to Materials by Design

## Local Python Environment Setup

*Separate setup instructions for macOS and Windows using WSL*

This environment is intended for work performed on your personal computer. It supports the Python-based scientific computing, data analysis, plotting, machine-learning, materials-science, and Jupyter activities used throughout the course.

- macOS: use the normal Terminal application.
- Windows: use Windows Subsystem for Linux (WSL) with an Ubuntu terminal.
- GPU support is not required for this local environment. A separate environment will be used later for HPC and GPU calculations.

**Choose one path:** If you use a Mac, follow Section A. If you use Windows, follow Section B. You do not need to complete both sections.

The course uses Miniforge, which provides the conda environment manager and uses the conda-forge package repository.

## Section A - macOS Setup

Complete this section from top to bottom if you are using a Mac. Run all commands in the normal macOS Terminal application.

### A1. Open Terminal

Open the Terminal application on your Mac.

### A2. Check that curl is available

Run the following command:

```
curl --version
```

If this prints information about curl, continue.

### A3. Download Miniforge

The following command automatically selects the correct macOS installer for your processor architecture:

```
curl -L -O "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-$(uname)-$(uname -m).sh"
```

Run the installer:

```
bash Miniforge3-$(uname)-$(uname -m).sh
```

In most cases, accept the default options. When asked whether Miniforge should initialize conda for your shell, answer:

```
yes
```

When installation is finished, close Terminal and open a new Terminal window.

### A4. Check that conda works

```
conda --version
```

You should see a conda version number. The exact version may differ.

### A5. Set conda-forge to strict channel priority

```
conda config --set channel_priority strict
```

You only need to do this once. It helps conda consistently obtain compatible packages from the same software ecosystem.

### A6. Create a course directory

Place the provided environment file, environment_local.yml, in a convenient directory. For example:

```
mkdir -p materials_by_design
cd materials_by_design
```

Copy or move environment_local.yml into this directory. Check that it is present:

```
ls
```

You should see environment_local.yml in the directory listing.

### A7. Create the course environment

```
conda env create -f environment_local.yml
```

Conda will download and install the required packages. The environment is named materials_by_design_local.

### A8. Activate the environment

```
conda activate materials_by_design_local
```

Your terminal prompt should now show the environment name near the beginning of the command line, for example:

```
(materials_by_design_local) student@computer:~$
```

### A9. Install PyTorch

With the course environment activated, run:

```
python -m pip install torch
```

### A10. Check the Python version

```
python --version
```

You should see Python 3.12.x.

### A11. Verify the environment

Place the provided verify_local_environment.py file in your working directory, then run:

```
python verify_local_environment.py
```

If everything is installed correctly, the final line should say:

```
Environment check completed successfully.
```

**macOS setup complete.** If you reached the successful environment check, skip Section B and continue to Section C.

## Section B - Windows Setup Using WSL + Ubuntu

Complete this section from top to bottom if you are using Windows. All commands below must be run inside your Ubuntu terminal through Windows Subsystem for Linux (WSL).

**Important:** Do not run these commands in Windows PowerShell or Command Prompt. For this course, the Python environment should live inside the Ubuntu/WSL environment.

### B1. Open the Ubuntu terminal

Launch your installed Ubuntu distribution through WSL. You should see a Linux-style command prompt.

### B2. Check that curl is available

```
curl --version
```

If curl is not installed, run:

```
sudo apt update
sudo apt install -y curl
```

### B3. Download Miniforge

Run the following command inside the Ubuntu/WSL terminal:

```
curl -L -O "https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-$(uname)-$(uname -m).sh"
```

Run the installer:

```
bash Miniforge3-$(uname)-$(uname -m).sh
```

In most cases, accept the default options. When asked whether Miniforge should initialize conda for your shell, answer:

```
yes
```

When installation is finished, close the Ubuntu terminal and open a new Ubuntu terminal.

### B4. Check that conda works

```
conda --version
```

You should see a conda version number. The exact version may differ.

### B5. Set conda-forge to strict channel priority

```
conda config --set channel_priority strict
```

You only need to do this once.

### B6. Create a course directory inside WSL

Create a course directory in your Ubuntu/WSL environment:

```
mkdir -p ~/materials_by_design
cd ~/materials_by_design
```

Place the provided environment_local.yml file in this directory. Then check that it is present:

```
ls
```

You should see environment_local.yml in the directory listing.

### B7. Create the course environment

```
conda env create -f environment_local.yml
```

Conda will download and install the required packages. The environment is named materials_by_design_local.

### B8. Activate the environment

```
conda activate materials_by_design_local
```

Your Ubuntu terminal prompt should now show the environment name near the beginning of the command line, for example:

```
(materials_by_design_local) student@computer:~$
```

### B9. Install the CPU version of PyTorch

GPU support is not required for the local course environment. With the course environment activated, run:

```
python -m pip install torch --index-url https://download.pytorch.org/whl/cpu
```

### B10. Check the Python version

```
python --version
```

You should see Python 3.12.x.

### B11. Verify the environment

Place the provided verify_local_environment.py file in your working directory, then run:

```
python verify_local_environment.py
```

If everything is installed correctly, the final line should say:

```
Environment check completed successfully.
```

**Windows/WSL setup complete.** If you reached the successful environment check, continue to Section C.

## Section C - Using the Course Environment

The commands in this section are the same for macOS and Windows/WSL unless otherwise noted.

### C1. Packages included in the environment

| Category | Major packages |
|---|---|
| Scientific computing | numpy, scipy |
| Data analysis | pandas, openpyxl, h5py |
| Plotting | matplotlib |
| Machine learning | scikit-learn, torch |
| Materials science | ase, pymatgen |
| Additional utilities | networkx, pyyaml, tqdm |
| Interactive Python | jupyterlab, ipykernel |

### C2. Running the introductory Python examples

First activate the environment:

```
conda activate materials_by_design_local
```

Move into the directory containing the Python scripts, for example:

```
cd python_basics
```

Run a script with:

```
python 00_python_basics_overview.py
```

Then continue through the examples:

```
python 01_logic_and_loops.py
python 02_lists_dictionaries_and_functions.py
```

### C3. Starting JupyterLab

```
conda activate materials_by_design_local
jupyter lab
```

JupyterLab should open in your web browser. Keep the terminal running while you are using JupyterLab.

### C4. Leaving the environment

```
conda deactivate
```

The environment name should disappear from the beginning of the terminal prompt.

### C5. Using the environment again later

You do not need to reinstall the environment every time you work on the course. Open the appropriate terminal for your computer and run:

```
conda activate materials_by_design_local
```

Then move to your course directory and run your scripts. When finished:

```
conda deactivate
```

### C6. Checking installed packages

To see all packages installed in the conda environment:

```
conda list
```

To check a particular package:

```
conda list numpy
conda list pymatgen
```

For packages installed using pip, you can also use:

```
python -m pip list
```

## Section D - Troubleshooting

### D1. Confirm that the course environment is active

```
conda activate materials_by_design_local
```

### D2. Check which Python executable is being used

```
which python
```

On macOS or Ubuntu/WSL, the returned path should contain something similar to:

```
miniforge3/envs/materials_by_design_local
```

### D3. Re-run the verification script

```
python verify_local_environment.py
```

Look for the first package that produces an error.

**Do not use the system Python.** Do not install missing course packages into the system Python installation. Install them only after activating materials_by_design_local.

### Quick reference

| Task | macOS | Windows |
|---|---|---|
| Open terminal | Terminal | Ubuntu terminal through WSL |
| Activate course environment | conda activate materials_by_design_local | conda activate materials_by_design_local |
| Install PyTorch | python -m pip install torch | python -m pip install torch --index-url https://download.pytorch.org/whl/cpu |
| Leave environment | conda deactivate | conda deactivate |